Poly(A) tail retention of LINE-1 transcripts under arsenite stress covaries with m^6A density and structural completeness

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Abstract

LINE-1 (L1) retrotransposons comprise 17% of the human genome and are pervasively transcribed across human cell types, yet how RNA modifications shape L1 transcript fate under cellular stress remains unknown. Here, we use Oxford Nanopore direct RNA sequencing across nine human cell lines (23 libraries) to simultaneously measure m^6A, poly(A) tail length, and transcript integrity on individual L1 molecules. L1 transcripts carry higher m^6A density than non-L1 transcripts, beyond sequence-composition expectation, and m^6A density varies with evolutionary age. Under arsenite stress, L1 poly(A) tails shorten while non-L1 transcripts remain unchanged. Young, full-length L1 retain their poly(A) tails, and overlapping sequence features—structural completeness, EN domain retention, high m^6A density, and long consensus span—each independently confer this protection when present in ancient elements. A composite score integrating these features separates stressed ancient L1 into subgroups spanning 66–137 nt in median poly(A). m^6A density covaries with poly(A) retention specifically under stress, with low-m^6A reads more frequently reaching critically short poly(A) lengths. Together, these findings reveal that m^6A density, uncorrelated with poly(A) under basal conditions, becomes a stress-specific correlate of poly(A) retention among structurally incomplete L1.

1 Introduction

Long interspersed nuclear element 1 (LINE-1, L1) retrotransposons are the most abundant autonomous mobile elements in the human genome, constituting approximately 17% of total genomic sequence [Bourque et al., 2018]. Although most of the ∼500,000 L1 copies are ancient, truncated remnants incapable of retrotransposition, a small number of evolutionary young, full-length elements (L1HS and L1PA subfamilies) retain retrotransposition competence [Lanciano and Cristofari, 2020]. Beyond parasitic replication, L1 transcripts serve regulatory functions, including nuclear scaffolds in embryonic stem cells [Percharde et al., 2018], long-range enhancers [Li et al., 2024a], and contributors to long noncoding RNA biogenesis [Kapusta et al., 2013]. Notably, ancient L1 fragments are preferentially exapted as cis-regulatory elements: transposable element–derived sequences contribute ∼25% of candidate cis-regulatory elements in the human genome, with LINE families among the largest contributors [Du et al., 2024]. Enhancer-associated copies retain higher sequence conservation and longer intact length than non-functional counterparts [Nishihara, 2019], and the probability of enhancer exaptation increases with element age [Simonti et al., 2017]. Thus, L1 RNA is both a potentially deleterious mobile element and a functional regulatory molecule. Multiple layers of host regulation—DNA methylation, histone modifications, and RNA-based mechanisms—act to restrain L1 activity [Deniz et al., 2019].

N^6-methyladenosine (m^6A) is the most prevalent internal modification of mammalian mRNA and a key regulator of L1 biology. m^6A has dual roles on L1 retrotransposons. METTL3-deposited m^6A in the L1 $5'$ UTR promotes cap-independent translation of ORF1p and ORF2p via eIF3 recruitment, enhancing retrotransposition [Hwang et al., 2021]. Hwang et al. further showed that m^6A consensus motifs in the L1 $5'$ UTR have been under positive evolutionary selection, indicating that L1 co-opts the host m^6A machinery. Conversely, the host cell exploits the same marks to suppress L1 activity—an m^6A-based innate defense mechanism that also targets other transposable elements [Chelmicki et al., 2021]—through multiple readers. The nuclear reader YTHDC1 recognizes m^6A-marked L1 RNA and recruits SETDB1 for H3K9me3 chromatin silencing [Liu et al., 2021], while the cytoplasmic reader YTHDF2 directs m^6A-L1 RNA to autophagy-mediated decay [Tian et al., 2024]. Beyond this antagonism, m^6A on intronic L1 participates in gene regulation, as SAFB engages m^6A-marked intronic L1 to modulate long gene expression [Xiong et al., 2021].

This range of m^{6}A-dependent functions has been the prevailing framework for understanding m^{6}A on L1.

However, several questions remain unaddressed. First, most studies focus on young L1 elements (L1HS, L1PA1–2). The m^{6}A landscape of *ancient* L1—approximately 86% of all L1 copies and the predominant source of exapted regulatory elements—remains largely unexplored, including whether the DRACH motif context required for METTL3 recognition is preserved despite millions of years of sequence divergence. Second, prior m^{6}A quantification of L1 has relied on short-read approaches (MeRIP-seq, miCLIP) with severe mapping ambiguity across the repetitive L1 family [Lanciano and Cristofari, 2020]; single-molecule m^{6}A quantification on L1 has not been achieved. Third, L1 m^{6}A behaviour under cellular stress and its relationship to L1 RNA stability have not been investigated, despite evidence that arsenite stress drives L1 ORF1p accumulation in stress granules [Goodier et al., 2007] and that m^{6}A mediates stress-granule triage of mRNAs [Anders et al., 2018].

Arsenite stress provides a useful context for studying L1 RNA fate. Sodium arsenite triggers rapid stress granule formation and activates a non-canonical RNA decay pathway, XRN1-mediated 5′ exonucleolytic digestion that leaves poly(A) tails intact [de Muro et al., 2024]. Canonical deadenylation is suppressed under arsenite, as the CCR4-NOT complex is sequestered into stress granules [Bhatt et al., 2025, preprint] and the deadenylase adaptors Tob and Pan3 are degraded [Funakoshi et al., 2015]. Whether non-translated L1 fragments—intronless, AU-rich RNA molecules that lack the ribosomal protection of young, actively translated elements—are subject to distinct decay kinetics under these conditions has not been examined.

Here, we leverage Oxford Nanopore direct RNA sequencing (DRS) across nine human cell lines to simultaneously measure m^{6}A, poly(A) tail length, and transcript integrity on individual L1 molecules. We report that L1 carries elevated m^{6}A density across evolutionary age, that arsenite selectively shortens L1 poly(A) tails while young L1 retain theirs, and that overlapping sequence features—including m^{6}A density—distinguish protected from vulnerable elements, revealing a stress-specific dimension of m^{6}A on L1 distinct from its established roles in retrotransposition and silencing.

2 Results

2.1 L1 RNA carries elevated m^{6}A density across human cell lines

We applied Oxford Nanopore direct RNA sequencing (DRS) to nine human cell lines and identified 54,234 L1 reads across 12,125 loci using a two-stage filter that removes ∼90% of host gene read-through (see Methods). Of these, 38,544 were non-embedded and 15,690 were embedded within host gene exons (Table S1). Ancient elements (L1MC, L1ME, L1M) comprised 93% and young elements (L1HS, L1PA1–3) 7% of the non-embedded reads (Supplementary Fig. S1). Long-read alignment largely resolves the multi-mapping ambiguity that confounds short-read L1 studies [Lanciano and Cristofari, 2020]: ancient L1 reads had a median mapping quality (MAPQ) of 60 with >80% uniquely mapped (Supplementary Table S1). Young L1 reads, owing to >97% inter-element sequence identity, had lower MAPQ (median 9–29 for L1HS–L1PA2), but the kilobase-scale read lengths of DRS still anchor most reads to individual loci—an advantage unavailable to 150-bp short reads. Because our primary conclusions rely on subfamily-aggregated metrics (m^{6}A/kb, poly(A) length) rather than locus-level quantification, the residual multi-mapping in young L1 does not affect the reported enrichments. Among non-embedded L1, 61% resided within introns and 39% in intergenic regions, with similar mapping quality and poly(A) distributions (Supplementary Table S1).

L1 transcripts carried m^{6}A at a density of 2.89 sites/kb, compared with 1.62 sites/kb for non-L1 transcripts (randomly subsampled reads from the same libraries, predominantly mRNA; see Methods), a 1.79-fold enrichment (Fig. 1a). This exceeds sequence-composition expectation, as L1 has only 1.22-fold more DRACH motifs per kb than non-L1 transcripts. The enrichment was consistent across all nine cell lines and persisted at every probability threshold tested , with high-confidence m^{6}A calls occurring more frequently on L1. m^{6}A density also correlated with evolutionary age. Young L1 (L1HS, L1PA1–3) exhibited the highest density (4.16 sites/kb), followed by ancient L1 (2.69 sites/kb) and non-L1 transcripts (1.62 sites/kb, Fig. 1a). The Young>Ancient gradient and L1/non-L1 enrichment were consistent across all nine cell lines (all $P < 10^{-7}$, Supplementary Fig. S4a), and subfamily-level m^{6}A/kb rankings were highly concordant between cell line pairs (median Spearman $\rho = 0.85$, Supplementary Fig. S4b), indicating sequence-intrinsic features. m^{6}A density showed modest positional variation along the L1 consensus, with ORF1-derived fragments

carrying the highest density.

Orthogonal validation with RNA004 chemistry and the Dorado basecaller reproduced the L1 enrichment. In length-matched comparisons, L1 reads carried 1.5–1.8-fold higher m^6A/kb than non-L1 reads (Supplementary Fig. S6), consistent with RNA002 results, and the Young>Ancient gradient was recapitulated (1.81-fold). L1 m^6A showed strong DRACH-motif specificity, with the DRACH motif hierarchy nearly identical between L1 and non-L1 (Spearman $\rho = 0.91$, $P = 1.7 \times 10^{-7}$), with GGACT ranking first in both. Among 539 position-reproducible m^6A sites on L1 ($\geq$50% of reads methylated at a given genomic position), 93% occurred at DRACH motifs (Supplementary Fig. S6).

To evaluate ancient L1, we asked whether the positional m^6A landscape mirrors that of young elements. When DRACH sites were mapped to the L1 consensus sequence, per-position methylation rates of ancient and young L1 were correlated (Spearman $\rho = 0.44$, $P = 1.3 \times 10^{-6}$, $n = 112$ consensus bins): positions that are m^6A hotspots in young L1 showed 1.78-fold higher methylation in ancient copies than non-hotspot positions ($P = 5.0 \times 10^{-94}$). Methylation rates at individual DRACH sites were predicted by flanking sequence conservation. Ancient L1 DRACH sites retaining Young-L1-like flanking sequences ($\leq$2 mismatches in an 11-mer window) were methylated at twice the rate of diverged sites (9.95% vs. 4.98%; $P = 1.3 \times 10^{-96}$), in line with reports that flanking sequence and local RNA structure influence methylation [Luo et al., 2022, Shachar et al., 2024]. The Young>Ancient gradient arises from two multiplicative factors. In length-matched reads (2–5 kb), young L1 carried 1.41-fold more DRACH motifs per kb and showed 1.64-fold higher per-DRACH methylation (Fisher $P < 10^{-30}$ for both). Together, these results indicate that L1 retrotransposons carry m^6A enrichment that exceeds sequence-compositional expectation, shows an evolutionary gradient, and displays consistent positional patterns across ancient elements.

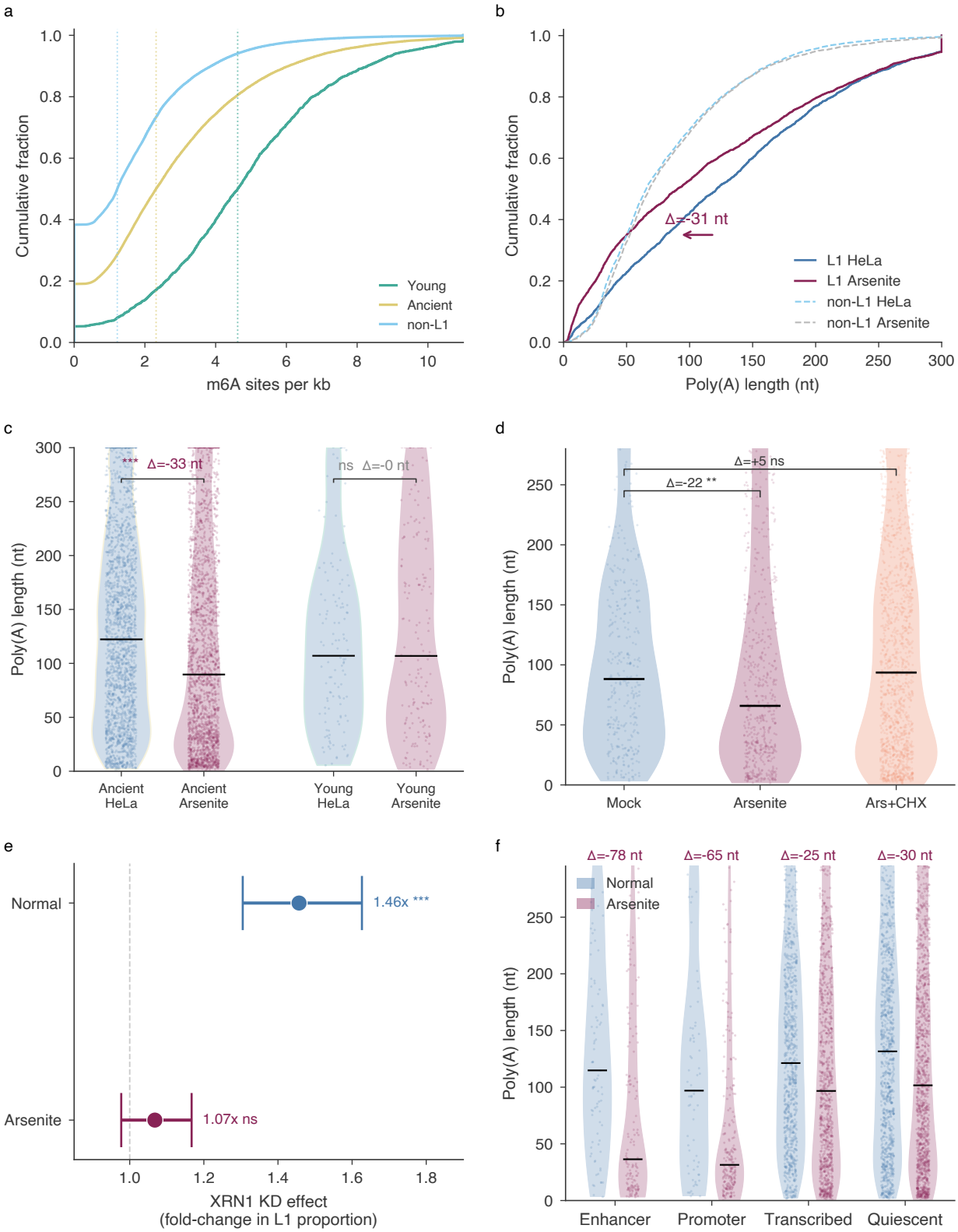

Figure 1: **L1 m^{6}A enrichment and arsenite-selective poly(A) shortening.** See Figure Legends for full description.

2.2 Arsenite triggers selective poly(A) shortening of non-embedded L1 transcripts

Having established that L1 carries elevated m^6A, we next asked how L1 transcripts behave under cellular stress. We analysed ONT DRS data from HeLa cells treated with 0.5 mM sodium arsenite for 60 minutes [de Muro et al., 2024] and compared poly(A) tail lengths between stressed and unstressed conditions. L1 poly(A) tails shortened by a median of 31 nt (122.5 to 91.5 nt, Mann–Whitney $P = 2.6 \times 10^{-17}$; Supplementary Fig. S1a), whereas non-L1 transcripts showed no significant change ($\Delta = +2.5$ nt, $P > 0.05$, Fig. 1b). The L1-specific effect ($\Delta\Delta = -33.5$ nt) indicates that shortening is restricted to L1 rather than a global change. To test whether this reflects a compositional shift toward loci with inherently shorter tails, we tracked 2,268 L1 loci detected exclusively under arsenite across nine unstressed cell lines. These loci had a median poly(A) of 77.5 nt under arsenite but 125.5 nt in other cell lines ($\Delta = -48.0$ nt, $P = 2.2 \times 10^{-35}$, Supplementary Fig. S1a), confirming a stress-dependent effect rather than locus-intrinsic properties. The shortening was comparable in intronic and intergenic L1 ($\Delta = -32.9$ and -27.1 nt, respectively; Supplementary Table S1), persisted at every read-length stratum, and was absent in annotated lncRNAs processed through identical pipelines ($\Delta = -2.9$ nt, Cohen's $d = -0.06$; Supplementary Fig. S7a–b), establishing it as L1-specific rather than a general feature of non-coding RNA.

Shortening was confined to ancient L1 elements (L1MC, L1ME, L1M, $\Delta = -33$ nt, $P = 1.7 \times 10^{-18}$), whereas young L1 were unaffected ($\Delta = -0.1$ nt, $P = 0.77$, Fig. 1c), even after matching for m^6A/kb. Within ancient subfamilies, vulnerability followed an evolutionary gradient: deeply ancient elements (L1M5, $\Delta = -49$ nt; L1ME4a, -37 nt; L1MC3, -64 nt) were most affected, whereas intermediate-age subfamilies (L1PA5–7) were unaffected ($P > 0.1$). Structural completeness was a strong determinant: full-length ancient elements were resistant ($\Delta = -9$ nt, ns; $n = 114$), whereas partial elements showed significant shortening ($\Delta = -33$ nt, $P = 1.2 \times 10^{-19}$; Supplementary Fig. S8b). Polyadenylation signal (PAS) strength also contributed: ancient L1 lacking any recognizable PAS showed stronger shortening ($\Delta = -35$ nt) than those with a PAS variant ($\Delta = -26$ nt; PAS $\times$ stress $P = 0.014$). Exon-embedded L1 within lncRNA or pseudogene exons (29% of total L1) were unaffected ($\Delta = -4.0$ nt, $P = 0.33$; Supplementary Fig. S2a).

Chromatin context further modulated the response among ancient, non-embedded L1. L1

in active regulatory chromatin (enhancers and active promoters; ChromHMM 15-state, HeLa-S3 E117) showed the strongest shortening ($\Delta = -73$ nt vs. -24 nt for non-regulatory L1, Fig. 1f; OLS stress×regulatory $\beta = -36.4$, $P = 1.1 \times 10^{-4}$), with enhancer-associated L1 most affected ($\Delta = -78$ nt, $P = 3.8 \times 10^{-7}$). This vulnerability was additive with m^{6}A, as regulatory L1 showed greater shortening across all m^{6}A quartiles (Supplementary Fig. S3a). Within stressed regulatory L1, the m^{6}A–poly(A) association was strong (Q4 median 59 nt vs. 28 nt for Q1, $P = 2.9 \times 10^{-4}$; Supplementary Fig. S3b). At the gene level, 69 regulatory L1 host genes with reads in both conditions showed a bidirectional response, and the m^{6}A/kb of each gene's L1 reads predicted the direction (Spearman $\rho = 0.38$, $P = 0.001$).

The shortening was genome-wide rather than driven by a few high-abundance loci, was independent of concurrent 5′ truncation at the single-molecule level ($r = 0.044$, $P = 0.12$).

To test whether shortening requires stress granule (SG) formation, we examined cycloheximide (CHX) co-treatment, which prevents SG assembly by trapping ribosomes on mRNAs. CHX rescued L1 poly(A) shortening: Median poly(A) under Ars+CHX was 93.6 nt, indistinguishable from mock (88.2 nt, $P = 0.66$), compared with 65.9 nt under arsenite alone ($P = 0.005$ vs. mock, Fig. 1d). The rescue was specific to ancient L1 ($\Delta = +29.3$ nt, $P = 1.3 \times 10^{-4}$) and uniform across m^{6}A quartiles , placing SG formation upstream of the m^{6}A-associated poly(A) variation described below. L1 RPM was 1.51-fold higher under Ars+CHX than arsenite alone (879 vs. 582).

We next assessed 5′ decay. XRN1 knockdown alone increased L1 proportion 1.46-fold ($P = 2.0 \times 10^{-11}$, χ^2 test), while XRN1 knockdown under arsenite had no additional effect (1.07-fold, $P = 0.15$), indicating convergence on the same pathway (Fig. 1e). L1 m^{6}A/kb remained constant across all four conditions (∼2.5/kb). Independent ribo-depleted Illumina RNA-seq under the same arsenite treatment (GSE278916, $n = 2$ per condition) showed reduced RNA abundance, with 53 of 67 ancient L1 subfamilies with significant changes showing downregulation (Supplementary Fig. S5).

2.3 m^{6}A density is associated with L1 poly(A) retention under stress

We next asked whether m^{6}A density is associated with the extent of poly(A) shortening on individual L1 molecules. Although m^{6}A density itself showed only a modest decrease under arsenite (2.50 vs. 2.62 sites/kb, 4.4% reduction, Mann–Whitney $P = 0.012$), the relationship between m^{6}A density and poly(A) length changed markedly. We stratified L1 reads into m^{6}A/kb quartiles and

compared poly(A) distributions (Fig. 2a). In unstressed HeLa cells, poly(A) tail length varied modestly across quartiles (Q1 = 109 nt, Q4 = 140 nt, Δ = 31 nt), whereas under arsenite the separation widened, with low-m^{6}A reads (Q1) showing a median poly(A) of 65 nt and high-m^{6}A reads (Q4) retaining 124 nt (Δ = 59 nt).

Under normal conditions, m^{6}A/kb and poly(A) were uncorrelated ($r \approx 0$; Supplementary Fig. S2b), whereas under arsenite a positive association emerged (per-read Spearman $\rho = 0.201$, $P = 3.3 \times 10^{-26}$; median Δpoly(A) from -42 nt in Q1 to -16 nt in Q4; Supplementary Fig. S6). PAS presence was an independent protective factor: when both PAS and m^{6}A were included in the same model, each remained significant (PAS $\times$ stress $P = 0.014$, m^{6}A/kb $\times$ stress $P = 1.4 \times 10^{-4}$). The coupling was driven by ancient L1, where the m^{6}A–poly(A) correlation increased from $r = 0.06$ (normal) to $r = 0.18$ ($P < 10^{-19}$) under stress, while young L1 showed no change ($r \approx 0.07$, ns; Fig. 2b). Stratifying stressed ancient L1 by m^{6}A and PAS status (CpG density averaged) revealed a 69-nt range in median poly(A), from 132 nt for high-m^{6}A with canonical PAS to 63 nt for low-m^{6}A without PAS (32% below 30 nt; Supplementary Fig. S6b).

Reads with critically short poly(A) tails (< 30 nt, below the PABPC1 dissociation threshold) were enriched among low-m^{6}A L1 under stress Among Q1 reads, 30.6% fell below this threshold, compared with 15.3% of Q4 reads (2.0-fold, Fisher's exact test, $P = 5.9 \times 10^{-11}$). Under normal conditions, this disparity was modest (Q1 14.6% vs. Q4 10.3%, $P = 0.036$). To estimate how much poly(A) variation is attributable to m^{6}A density, we aggregated reads by genomic L1 locus (median m^{6}A/kb and median poly(A) per locus; loci with $\geq$5 reads). The per-locus Spearman correlation was $\rho = 0.51$ ($P = 1.7 \times 10^{-4}$, $n = 49$ loci), with an OLS $R^2 = 0.12$ and a weighted least-squares $R^2 = 0.18$ (Supplementary Table S1). The effect was absent in unstressed loci ($R^2 = 0.003$, $n = 140$), confirming stress dependence. Together, these data indicate that L1 poly(A) retention under stress is associated with a hierarchy of molecular features: PAS strength as the primary constitutive factor, m^{6}A density as a stress-specific correlate, and chromatin context as an independent modulator.

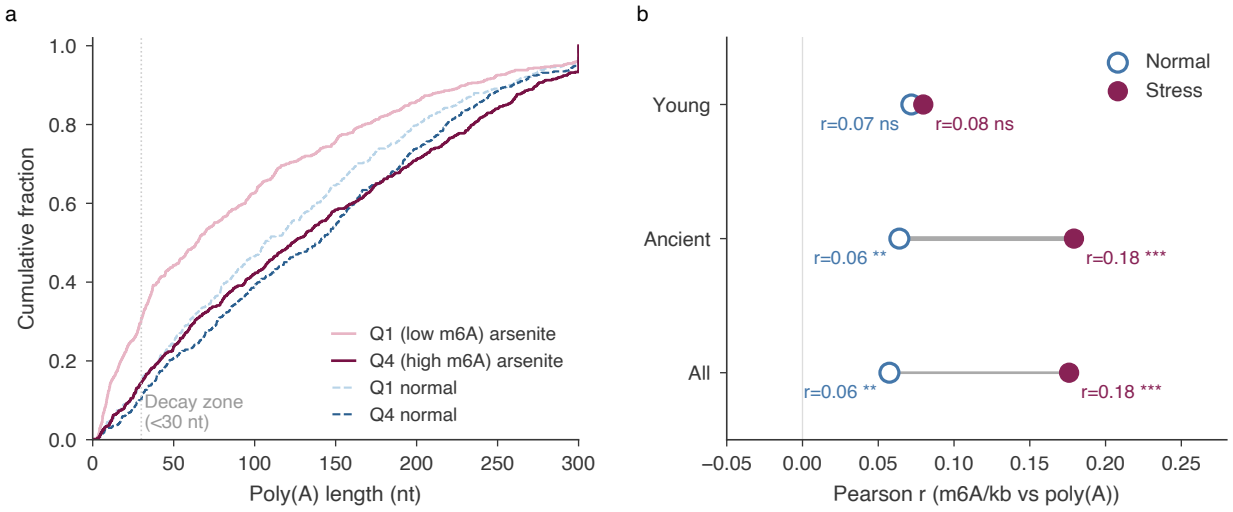

Figure 2: **m^{6}A density is associated with L1 poly(A) retention under stress.** See Figure Legends for full description.

2.4 Sequence features of young L1 confer stress immunity

Having established that m^{6}A density covaries with poly(A) retention under stress, we asked what sequence-encoded features distinguish young L1, which are immune to shortening, from vulnerable ancient elements. Young L1 differed from ancient elements across five axes: m^{6}A/kb (1.8-fold), full-length fraction (19.9-fold), EN domain retention (4.0-fold), consensus span (8.8-fold), and 3′ UTR coverage (1.6-fold; Supplementary Fig. S8a). Each of these features independently conferred stress immunity when present in ancient L1: full-length ancient elements ($\Delta = -9$ nt, ns), EN-containing elements ($\Delta = -6.1$ nt, ns), elements with high m^{6}A ($\geq$ young median; $\Delta = -12.6$ nt, ns), and elements with long consensus span ($\Delta = +5.6$ nt, ns) were all resistant, whereas ancient elements lacking each feature showed significant shortening ($\Delta = -34$ to -47 nt, all $P < 0.001$; Supplementary Fig. S8b). The sole exception was 3′ UTR coverage, which did not protect ancient elements ($\Delta = -24.4$ nt, $P = 4.3 \times 10^{-7}$), consistent with the requirement for upstream structural features beyond the PAS. A composite immunity score (0–3, one point each for full-length status, EN domain retention, and high m^{6}A/kb) showed a dose-dependent relationship with poly(A) retention under stress: score 0 median 66 nt, score 1 91 nt, score 2 132 nt, score 3+ 137 nt (Spearman $\rho = 0.169$, $P = 5.5 \times 10^{-18}$; Supplementary Fig. S8c).

At the motif level, the EN domain harbors the highest DRACH density in L1 (43.2/kb, 1.38-fold the consensus average) alongside near-complete CpG depletion (2.8/kb, obs/exp = 0.08; Supplementary Fig. S8d), creating a region maximally accessible to METTL3-mediated m^6A deposition while invisible to ZAP-mediated CpG surveillance [Takata et al., 2017, Moldovan and Moran, 2015]. Consistent with this, mutation sensitivity mapping showed that stress-sensitive ancient L1 carry significantly higher mutation rates specifically in the EN domain . m^6A/kb retained independent predictive value after controlling for structural features. Among stressed ancient L1 reads (n = 2,572), a likelihood ratio test comparing OLS models with and without m^6A/kb (controlling for full-length status, EN domain, consensus span, and 3′ UTR coverage) yielded $\chi^2 = 57.9$, $P = 2.7 \times 10^{-14}$, with each unit increase in m^6A/kb associated with +6.3 nt longer poly(A) ($\Delta R^2 = 2.2\%$). The larger coefficient compared with the pooled OLS ($\beta = 4.2$, Supplementary Table S2) reflects the restriction to stressed ancient elements, where the m^6A–poly(A) association is strongest. These results place m^6A density alongside structural completeness and PAS strength in a multi-feature framework that quantitatively predicts L1 poly(A) fate under stress.

3 Discussion

We show that young, full-length L1 retain their poly(A) tails under arsenite stress, and that overlapping sequence features—structural completeness, EN domain retention, high m^6A density, and long consensus span—each independently confer this protection. Ancient L1 lacking these features undergo selective poly(A) shortening, and among these vulnerable elements, m^6A density covaries with poly(A) retention in a dose-dependent pattern. The m^6A marks that promote retrotransposition [Hwang et al., 2021] and mediate transcriptional silencing [Liu et al., 2021, Tian et al., 2024] thus show a third, stress-dependent dimension: an association with differential poly(A) retention among structurally incomplete L1.

L1 poly(A) shortening under arsenite is unexpected given that canonical deadenylation is suppressed: CCR4-NOT is sequestered into stress granules [Bhatt et al., 2025, preprint] and the deadenylase adaptors Tob and Pan3 are degraded [Funakoshi et al., 2015]. Non-L1 transcripts accordingly retain their poly(A) (Fig. 1b), making L1 a notable exception. Although retrotransposon poly(A) tails can form protective triple-helix structures in plant systems [Li et al., 2025], human

L1 lacks the ENE motif required for such structures, and its poly(A) tails are clearly vulnerable to shortening. L1, as non-translated, intronless, AU-rich RNA, may instead engage non-canonical decay. The HUSH–NEXT axis targets intronless transcripts for nuclear decay [Tchasovnikarova et al., 2022, Garland et al., 2022], which would explain the resistance of exon-embedded L1 that inherit spliced architecture. MOV10–DCP2 mediates 5′ decay through phase-separated granules [Li et al., 2023], a mechanism that may underlie the concurrent 5′ truncation we observe. Cycloheximide blocks stress granule assembly and rescues poly(A) (Fig. 1d), placing shortening downstream of SG formation, as reported for SG-dependent 5′ decay [de Muro et al., 2024].

L1 additionally undergoes 3′ poly(A) shortening, a process distinct from 5′ decay: the two are independent at the single-molecule level, and XRN1—which can process m^6A-containing RNA without impediment [Athapattu et al., 2021]—degrades L1 non-selectively (L1 m^6A/kb unchanged across all four factorial conditions). In contrast, poly(A) shortening correlates with m^6A density and is strongest in regulatory chromatin (Fig. 1f). These axes separate bulk turnover (XRN1) from stress-inducible poly(A) shortening that is associated with m^6A density. Supporting this dichotomy, L1 reads experiencing 3′-only vulnerability carried lower m^6A/kb than reads affected by neither pathway (1.42 vs. 2.04, $P = 1.0 \times 10^{-15}$), whereas XRN1 sensitivity was uniform across m^6A strata. The coupling was similar in intronic and intergenic L1 ($r \approx 0.20$ for both), indicating that selectivity reflects transcript-intrinsic features rather than genomic context.

How might a modification associated with decay under basal conditions become associated with retention under stress? Arsenite relocates YTHDF1/3 to SG peripheries and recruits YTHDF2 for structural maintenance rather than decay [Fu and Zhuang, 2020, Li et al., 2024b], potentially allowing IGF2BP stability-promoting readers [Huang et al., 2018] to dominate. However, L1 may be a poor YTHDF2 substrate given its lack of translational context, and YTHDF2 phase separation under arsenite [Wang et al., 2023] would predict enhanced decay rather than protection—distinguishing these models will require reader-specific perturbation. The m^6A–poly(A) association on L1 inverts the canonical mRNA relationship, where m^6A promotes cotranslational decay via CCR4-NOT recruitment [Lima et al., 2017, Absmeier et al., 2024]. Ancient L1, presumably untranslated, escape this pathway, which may explain their longer baseline poly(A) (122 vs. 107 nt) but also deprive them of ribosome-mediated protection under stress.

Young L1 stress immunity reflects overlapping sequence-encoded protections rather than a single

determinant (Supplementary Fig. S8). As domains are lost through 5′ truncation, these protections are progressively stripped, and m^6A emerges as a late-stage correlate whose association with poly(A) retention becomes detectable only in structurally incomplete elements that lack other protections. This framing aligns with the m^6A surveillance model, which proposes that m^6A preferentially marks intronless transcripts as quality-control targets [Ke, 2026]. Under basal conditions, such marking may facilitate decay; under stress, when SG sequestration disables canonical decay, the same marks may be co-opted for differential poly(A) retention.

These observations have implications for L1 regulatory biology. Supporting a functional role for the m^6A enrichment we report, m^6A on ancient L1PA elements directly regulates EP300/KAP1 binding and LTR trans-silencing in human embryonic stem cells [Zhu et al., 2025]. The selective poly(A) shortening we observe under stress could therefore alter the composition of the regulatory L1 RNA pool, although demonstrating downstream chromatin consequences will require perturbation of m^6A writers specifically on L1.

The stress resilience of young L1 may help explain a long-observed phenomenon: L1 RNA accumulation under cellular stress and senescence [De Cecco et al., 2019, Gorbunova et al., 2021]. Prior studies have attributed this accumulation to transcriptional derepression—DNA demethylation and chromatin relaxation unlock L1 promoters [Gorbunova et al., 2021]. Our data suggest that post-transcriptional stability is a complementary mechanism. Young, full-length L1 carry overlapping protections (Supplementary Fig. S8a,b) that shield their poly(A) tails from stress-induced shortening, allowing newly transcribed L1 RNA to persist. In contrast, ancient L1 fragments lacking these features undergo selective poly(A) shortening and decay. The net effect—young L1 RNA accumulation alongside ancient L1 RNA depletion—may reshape the cellular L1 RNA pool under stress, favoring retrotransposition-competent over regulatory transcripts. These features are sequence-encoded and cross-cell-line consistent (CV = 0.112 for m^6A/kb), offering a quantitative basis for predicting which L1 elements are most susceptible to stress-induced poly(A) shortening.

Several limitations constrain interpretation. The m^6A–poly(A) association, while robust and dose-dependent, remains correlational. The per-read OLS model explains a modest fraction of poly(A) variance ($R^2 = 0.031$ for stressed reads; $R^2 = 0.039$ pooled, Supplementary Table S2), as expected given single-molecule noise and the multiple determinants of poly(A) length. Per-locus aggregation raises the explained variance to $R^2 = 0.12$–0.18 (Supplementary Table S1), but rests on

only 49 loci with ≥5 reads (bootstrap 95% CI for ρ: 0.28–0.70, 10,000 resamples). The association is significant across all five probability thresholds tested (50% to 90%, all $P < 2\times10^{-4}$; Supplementary Table S3), arguing against threshold-dependent circularity. Perturbation of m^6A writers under stress would be needed to establish causality. We note that analysis of a published METTL3 knockout dataset (HEK293T; PRJEB40872) showed no significant reduction in L1 m^6A/kb (fold change = 0.97, $P > 0.05$), though non-L1 control transcripts showed a modest decrease (0.91-fold, $P = 0.035$). This null result may reflect cell-type differences, residual METTL3 activity, or limitations of modification calling on repetitive sequences, but it underscores the need for L1-targeted writer perturbation under stress conditions.

DRS captures only surviving polyadenylated RNA and cannot measure decay rates directly. The 3′ bias of DRS (90% of reads within 1 kb of the 3′ end) limits 5′ UTR coverage; the median L1 body overlap per read is 0.35, so read-level m^6A/kb includes flanking modifications, though restricting to high-overlap reads preserves the dose-response pattern. ChromHMM states from E117 (HeLa-S3) were used as a proxy for HeLa chromatin; subline-specific epigenetic divergence cannot be excluded. All stress-related findings derive from a single cell line (HeLa) treated with arsenite for 60 minutes. Baseline m^6A enrichment is consistent across nine cell lines (CV = 0.112), but generalizability to other cell types and stressors remains to be established. MAFIA was trained on non-repetitive transcripts [Pratanwanich et al., 2021, Hendra et al., 2022], and nanopore modification callers show variable performance across sequence contexts [Deng et al., 2024]; benchmarking specifically on repetitive elements remains a priority. However, orthogonal RNA004/Dorado validation reproduced the L1 enrichment with 93% DRACH motif specificity (Supplementary Fig. S6), supporting modification call reliability. Time-course and multi-cell-line stress experiments are important next steps. For the XRN1 KD experiments, statistical comparisons are based on read-level variation within single biological replicates per condition due to a sequencing-platform batch effect (see Methods).

4 Materials and Methods

Data sources

We reanalysed publicly available Oxford Nanopore direct RNA sequencing (DRS) data from three sources. **(i)** Baseline transcriptome data for nine human cell lines (A549, H9, HCT116, HeLa, HepG2, HEYA8, K562, MCF7, and SH-SY5Y; 23 libraries total) were obtained from the SG-NEx project [Chen et al., 2025]. All libraries used the ONT SQK-RNA002 kit (MinION, FLO-MIN106, R9.4.1 chemistry) with at least two biological replicates per cell line. **(ii)** HeLa arsenite-stress data (0.5 mM $NaAsO_2$, 60 min; three replicates), cycloheximide co-treatment data, and XRN1 knockdown data were obtained from PRJNA842344 [de Muro et al., 2024]. For XRN1 KD/CHX, replicate 1 (PromethION) and replicate 2 (MinION) showed a 2.94-fold platform batch effect; only MinION replicate 2 was used for all four factorial conditions (mock, XRN1 KD, arsenite, arsenite + XRN1 KD), so comparisons are based on read-level variation within single replicates. MCF7-EV data were from the same SG-NEx panel. **(iii)** For RNA004 validation, HeLa DRS data were basecalled with Dorado v0.8.0 (ONT; model `rna004_130bps_sup@v5.2.0_inosine_m6A_2OmeA@v1`), which detects m^6A at all adenosine positions regardless of motif context. Per-read m^6A/kb was computed from the BAM-encoded per-base modification probabilities (threshold ≥ 0.80) to compare with MAFIA-derived values. Raw signal data were obtained as FAST5 or POD5 files.

Basecalling and alignment

Raw FAST5 files were basecalled using Guppy v6.0.0 (ONT) with the high-accuracy RNA model (`rna_r9.4.1_70bps_hac`). Basecalled reads were aligned to GRCh38/hg38 using minimap2 v2.28 [Li, 2018] with splice-aware settings and a GENCODE v38 junction BED file (`-junc-bed`). Secondary alignments were discarded and BAM files were sorted and indexed with samtools v1.23.

Two-stage L1 identification filter

L1 reads were identified through a two-stage filter designed to eliminate host gene read-through contamination.

Stage 1 (coarse filter). Aligned reads were intersected with RepeatMasker L1 annotations (UCSC rmsk track, hg38) using bedtools v2.31.0. Reads with $\geq$10% overlap to annotated L1

elements were retained.

Stage 2 (fine filter). Stage 1 reads were filtered by four criteria: (i) exclusion of reads with splice junctions (CIGAR 'N' operations > 50 bp), (ii) selection of the best alignment per read, (iii) exclusion of reads with ≥100 bp overlap to protein-coding exons (GENCODE v38), and (iv) strand concordance with the annotated L1 element. Reads passing both stages were classified as non-embedded L1 ("PASS" in the analysis pipeline). The 100 bp exon overlap threshold was empirically validated: 92.1% of non-embedded L1 reads have zero exon overlap, and the biological signal (arsenite poly(A) shortening) is indistinguishable between zero-overlap and near-threshold reads . Reads failing Stage 2 criterion (iii) due to overlap with long noncoding RNA or pseudogene exons (but not protein-coding exons) were classified as exon-embedded L1.

Non-L1 reads from the same libraries were randomly subsampled to form the comparison group and processed through identical nanopolish and MAFIA pipelines. Because DRS captures poly(A)$^+$ RNA, these non-L1 reads are dominated by highly expressed protein-coding transcripts: 93.9% protein-coding mRNA, 3.2% MT-rRNA, and 1.7% lncRNA (n = 40,659 reads across 6,512 genes). The top contributors include *MT-RNR2*, *EEF1A1*, *FTL*, and ribosomal protein mRNAs. This composition differs from L1 reads in sequence context, expression level, and transcript architecture. To address potential bias from expression-level differences in m^{6}A/kb, we performed expression-matched analyses using gene-weighted and quintile-stratified controls (see Methodological considerations). These reduced the L1/non-L1 m^{6}A/kb ratio but did not alter the principal conclusions.

As an additional control for non-coding RNA identity, annotated lncRNAs (GENCODE v38 biotype "lncRNA") were independently identified from the same libraries and processed through identical nanopolish poly(A) and MAFIA pipelines (6 libraries × ∼3,000–4,800 MAFIA reads). Read-length-matched analyses confirmed that poly(A) and m^{6}A differences were not attributable to transcript length (Supplementary Fig. S7).

L1 subfamily and age classification

L1 subfamily assignments were derived from RepeatMasker annotations. Elements were classified as "young" (L1HS, L1PA1, L1PA2, L1PA3) or "ancient" (all other L1 subfamilies including L1MC, L1ME, L1M families). Consensus positions within the full-length L1 reference were obtained from RepeatMasker `repStart`, `repEnd`, and `repLeft` fields (UCSC rmsk track). Functional domains

were defined on the L1HS consensus (6,064 bp): 5′ UTR (1–908), ORF1 (909–1,990), ORF2 (1,991–5,817), and 3′ UTR (5,818–6,064). Each genomic L1 element was assigned a primary domain based on the largest consensus overlap, and elements spanning all four domains were classified as "full-length". For the ORF2 sub-domain analysis, the EN domain was defined as consensus positions 1,991–2,708, the RT domain as 2,709–4,149, and the C-rich domain as 4,150–5,817.

Mutation sensitivity mapping

To identify L1 consensus positions where mutations correlate with stress vulnerability, we extracted the genomic sequences of 508 ancient L1 loci (top and bottom poly(A) quartiles under arsenite) using bedtools getfasta (strand-aware). Sequences were aligned to the L1HS consensus (GenBank L19088.1) with minimap2 (`--eqx --MD`). Per-position match/mismatch counts were tallied separately for stress-sensitive and stress-resistant loci. Mutation rate differences were assessed in 100-bp sliding windows (50-bp step) using Fisher's exact test with Bonferroni correction. Only windows with $\geq$100 aligned bases in both groups were included.

Poly(A) tail length estimation

Poly(A) tail lengths were estimated from raw nanopore signal using nanopolish v0.14.0 [Simpson et al., 2017] with the `polya` module, which fits a piecewise model to the poly(A) signal segment. Only reads with a "PASS" quality flag from nanopolish were retained.

m^{6}A modification calling

m^{6}A modifications were called from aligned BAM files using psi-co-mAFiA (v0.0.1; Hendra et al., 2022), which assigns per-base m^{6}A modification probabilities (ChEBI:21891) encoded as MM/ML tags in the BAM output.

Modification calls were parsed from MM/ML BAM tags, matching both forward-strand (N+21891) and reverse-strand (N−21891) tags to capture modifications on all mapped reads. A probability threshold of ≥ 0.80 (ML $\geq$ 204/255) was used to classify sites as modification-positive for all analyses.

m^6A density and decomposition analysis

m^6A density was computed as the number of modification-positive DRACH sites per kilobase of aligned read length. To determine whether L1 m^6A density exceeds what would be expected from elevated DRACH motif frequency in AT-rich L1 sequences, we counted DRACH motif positions (D = A/G/T, R = A/G, H = A/C/T) within each read's aligned reference sequence. Reference sequences were extracted in a CIGAR-aware manner, walking only through M (match), = (sequence match), and X (mismatch) CIGAR operations to exclude intronic regions in spliced non-L1 reads. This CIGAR-aware extraction was critical for non-L1 transcripts, where using `reference_end` − `reference_start` (which includes intronic spans) inflated motif counts by approximately 4-fold.

Statistical analyses

All statistical analyses were performed in Python 3.10 using scipy, statsmodels, and pandas. Two-group comparisons used the Mann–Whitney U test (poly(A) tail lengths, m^6A/kb) or Fisher's exact test (m^6A site counts). Multi-group comparisons used the Kruskal–Wallis test. Effect sizes for modification enrichment were computed as fold change of m^6A/kb. The relationship between m^6A density and poly(A) tail length was modeled using ordinary least-squares (OLS) regression with the model:

$$\text{poly(A)} = \beta_0 + \beta_1 \cdot \text{m}^6\text{A/kb} + \beta_2 \cdot \text{stress} + \beta_3 \cdot (\text{m}^6\text{A/kb} \times \text{stress}) + \epsilon \tag{1}$$

where stress is a binary indicator for the arsenite condition. The full model included additional covariates for read length (z-scored), L1 age (young vs. ancient), and their interactions (Supplementary Table S2). The interaction coefficient β_3 tests whether the m^6A–poly(A) relationship differs between stressed and unstressed conditions. Pearson and Spearman correlation coefficients were computed for stratified subgroups (L1 age, genomic context, stress condition).

To reduce single-molecule measurement noise, we performed per-locus aggregation by grouping reads that mapped to the same genomic L1 insertion (defined by chromosome, start, end, and strand). For each locus, median m^6A/kb and median poly(A) length were computed. Per-locus regression was performed using both ordinary least-squares and weighted least-squares (weight = number of reads per locus) on loci with a minimum read depth of 3 or 5 reads. Multiple testing

correction was not applied to the primary analyses, as each test addressed a distinct, pre-specified hypothesis. Where exploratory analyses involved multiple comparisons, this is noted in the text.

Illumina RNA-seq validation

To orthogonally validate L1 expression changes under arsenite, we analysed publicly available ribo-depleted Illumina RNA-seq data from HeLa cells (GSE278916, untreated $n = 2$, arsenite $n = 2$). Reads were aligned to the human genome (hg38, GENCODE v38) using STAR v2.7.11a with multi-mapper settings compatible with TE quantification (`-outFilterMultimapNmax 100`). L1 subfamily-level expression was quantified with TEtranscripts v2.2.3 using the EM algorithm for multi-mapper resolution and the RepeatMasker TE annotation (hg38_rmsk_TE.gtf). Differential expression was assessed with DESeq2 (integrated within TEtranscripts). As a complementary approach, element-level L1 counts were obtained with featureCounts (Subread v2.0.6) in fractional multi-mapper mode and normalized using DESeq2-style median-of-ratios gene-level size factors.

Gene ontology and genomic context analysis

Host genes harboring intronic L1 were identified by intersecting L1 genomic coordinates with GENCODE v38 gene annotations. Gene Ontology enrichment was performed using Enrichr. PAS motifs were scanned within 50 bp upstream of each L1 element's $3'$ end in the reference genome, searching for the canonical AATAAA and 11 variant PAS hexamers.

ChromHMM chromatin state annotation

Chromatin state annotations were obtained from the Roadmap Epigenomics 15-state ChromHMM model [Roadmap Epigenomics Consortium et al., 2015]. For HeLa analyses, the E117 (HeLa-S3) segmentation BED file was used. Each L1 element was assigned to a chromatin state by intersecting its genomic midpoint with the ChromHMM state annotation using bedtools intersect. States were grouped into functional categories: "regulatory" (active enhancers [states 6–7] and active promoters [states 1–3]), "transcribed" (states 4–5, 9–11), and "quiescent/repressed" (states 12–15).

5 Data Availability

This study reanalysed publicly available datasets; no new sequencing data were generated. Baseline DRS data for nine human cell lines were obtained from the SG-NEx project (ENA: ERP125816). Arsenite-stress, cycloheximide, and XRN1 KD data were from PRJNA842344 [de Muro et al., 2024]. PUS enzyme knockdown data were from PRJNA1220613 (BE(2)-C) and PRJNA1092333 (SH-SY5Y). Analysis code and figure-generation scripts are available at `https://github.com/junsoopablo/L1-m6A-polyA`.

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Author Contributions

J.K. conceived and designed the study, performed all computational analyses, and wrote the manuscript. S.L. acquired and curated the public datasets. S.H. contributed to data interpretation. All authors reviewed the manuscript.

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Figure Legends

Figure 1. L1 m^{6}A enrichment and arsenite-selective poly(A) shortening.

(a) ECDF of per-read m^{6}A/kb for young L1 (L1HS, L1PA1–3), ancient L1, and non-L1 transcripts. Dotted vertical lines mark group medians. L1 curves are right-shifted, indicating higher m^{6}A density (1.79-fold enrichment). **(b)** ECDF of poly(A) tail length for L1 and non-L1 transcripts in unstressed HeLa versus arsenite-treated HeLa. The L1 curve shifts leftward under arsenite ($\Delta = -31$ nt, $P = 2.6 \times 10^{-17}$) while non-L1 curves remain unchanged. **(c)** Poly(A) distributions for ancient and young L1 under normal and arsenite conditions. Ancient L1 show shortening ($\Delta = -33$ nt, $P = 1.7 \times 10^{-18}$) while young L1 are unaffected ($\Delta = -0.1$ nt, ns). **(d)** Cycloheximide (CHX) rescue. Ars+CHX poly(A) is indistinguishable from mock ($P = 0.66$), placing shortening downstream of stress granule formation. **(e)** XRN1 knockdown increases L1 proportion 1.46-fold under normal conditions ($P = 2.0 \times 10^{-11}$) but has no additional effect under arsenite (1.07-fold, $P = 0.15$), indicating pathway convergence. Error bars: 95% CI. **(f)** Chromatin state modulates arsenite vulnerability. L1 in regulatory chromatin show the strongest shortening ($\Delta = -73$ nt), while transcribed and quiescent regions show moderate effects. ChromHMM 15-state, HeLa-S3 E117.

Figure 2. m^{6}A density is associated with L1 poly(A) retention under stress.

(a) ECDF of poly(A) length stratified by m^{6}A/kb quartile. Under stress, low-m^{6}A reads (Q1) shift leftward while high-m^{6}A reads (Q4) retain poly(A). Dotted line: 30-nt PABP dissociation threshold (Q1/Q4 ratio 2.0-fold, $P = 5.9 \times 10^{-11}$). **(b)** Pearson r (m^{6}A/kb vs. poly(A)) under normal versus stress for young L1, ancient L1, and all L1. The correlation emerges under stress in ancient L1 ($r = 0.06 \rightarrow 0.18$, $P < 10^{-19}$), while young L1 shows no change.